Task 2: Analysis & HCI Reflection

Human Perception & Performance Tester | Dziwornu Edith | 21151366 | Human-Computer Interaction

Method. My roommate (participant P01, not the developer) completed 8 identification trials and three target-hitting blocks of 18 clicks (54 trials). Contrast, size and brightness were changed with the app's controls and every trial was logged automatically to the CSV file. With one participant the results are indicative only; I note where practice or the fixed block order may have affected them.

a. Human limits

Identification. Practice affected these times more than any visual setting: the first trial took 15.7 s, while later High-contrast trials took 5.2, 5.3 and 1.3 s. The clearest condition effect came from low contrast. The Necker cube at Low contrast (1.9:1, Large size) took 4.05 s and was rated 1/5 for ease, the only 1 given, about three times slower than the same figure at High contrast (21:1) a few trials earlier (1.30 s). Size also changed in that trial, so I treat it as suggestive, not proof. The two dimmed trials (51% and 57% brightness) were answered quickly (1.46 s and 1.13 s) but both were rated only 2/5, and they came after the participant had practised.

Target hitting. Dimming the screen to 49% made targets harder to acquire: mean time rose from 1.44 s to 1.62 s (+12%), misses per trial rose from 0.05 to 0.28, and the ease rating fell from 5 to 2. Low contrast did not slow the clicks (1.35 s) but lowered the ease rating to 3, so the user worked harder for the same speed. The targets were large filled circles, which remain detectable at 1.9:1, and the blocks were not counterbalanced, so practice may have hidden a slowdown.

Target-hitting block (18 trials each)	Mean time (s)	Misses / trial	Ease (1-5)
High contrast, 100% brightness	1.44	0.05	5
Low contrast (1.9:1), 100% brightness	1.35	0.11	3
High contrast, 49% brightness	1.62	0.28	2

These limits come from the visual system: contrast sensitivity falls as the luminance difference shrinks, acuity depends on the visual angle an object covers, and a dim display reduces the usable luminance difference. They explain familiar failures. Early ATM screens were small, dim and low-contrast, and sunlight glare made prompts easy to misread. Many websites use light-grey text on white below the WCAG minimum of 4.5:1, which slows reading. My ratings fell further than my times, which shows why timing alone underestimates the cost: users compensate with effort and frustration.

b. Applying Fitts' Law

Fitts' Law predicts $MT = a + b \log_2(D/W + 1)$: larger targets (W) and shorter distances (D) are faster.

Target diameter	20 px	45 px	90 px	Distance	100 px	200 px	320 px
Mean time (s)	1.81	1.39	1.20	Mean time (s)	1.44	1.49	1.47
Misses / trial	0.39	0.00	0.06	Misses / trial	0.11	0.22	0.11

Larger targets clearly reduced movement time (34% faster from 20 px to 90 px) and small targets caused far more misses. **Shorter distances did not:** the three distances differ by only 0.03 s. A straight-line fit over all 54 trials gave $MT = 0.89 + 0.24 \log_2(D/W + 1)$ seconds, with $r = 0.34$.

Does this match Fitts' Law? Partly. The direction is right and the size is realistic: a slope of 0.24 s/bit predicts about 0.7 s between the easiest and hardest combinations (1.1 to 4.1 bits), and I observed 0.74 s (1.13 s for 100 px distance with a 90 px target, 1.87 s for 320 px with a 20 px target). The weak correlation and flat distance effect do not match. Likely reasons: distances covered only 100 to 320 px, so extra distance added little compared with the aiming and reaction time in every measurement; my design changed width 4.5 times but distance only 3.2 times, so width dominated the difficulty; and each combination was repeated just twice by one participant. A wider range of distances and more repetitions would test the distance term properly.

c. Design for diversity

Colour-blind user. My conditions varied luminance (black to grey on white), which colour-vision deficiency barely changes, so I would expect similar performance here. The risk is any design that carries meaning by colour alone, such as red/green status messages or coloured chart lines, which about 1 in 12 men cannot separate. Adding labels, icons or patterns gives a redundant cue. **Elderly user with reduced acuity.** Ageing lowers acuity and contrast sensitivity and often slows and unsteadies the hand. My participant, with normal vision, already rated dim and low-contrast conditions 2 to 3 out of 5 and missed small targets most (0.39 misses per trial), so I would expect larger penalties and a steeper Fitts slope for an older user. The remedies are high contrast, larger text and large, well-spaced targets of at least 44 px, with no need for precise or timed movements.

d. Recognition vs recall

The interface keeps information on screen so users never have to remember steps. Identification answers are visible buttons (Vase / Faces / Both / Unsure), so the participant picks rather than types or recalls a word. The Fitts task shows a labelled START circle, a "Trial n of 18" counter and an instruction on every trial. The 1 to 5 rating buttons carry their meaning ("1 = very difficult ... 5 = very easy") each time. For the experimenter, the panel always shows the

current condition and contrast ratio, a preview of sample text shows the effect of each setting, and controls lock during a task so settings cannot be changed or forgotten by accident.

Limitations

One participant, 8 identification trials spread across conditions (some changed two settings at once), and fixed block order. A stronger study would test several people, counterbalance the order of conditions, change one setting at a time and repeat each combination more often.